

Create ProjectDB SQLite Database

Assignment 08.1: SQL Queries

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```
library(RSQLite)
```

```
## Warning: package 'RSQLite' was built under R version 4.4.3
```

```
fpath = ""  
dbfile = "projectdb3.db"  
  
# if database file already exists, we connect to it, otherwise  
# we create a new database  
dbcon <- dbConnect(RSQLite::SQLite(), paste0(fpath,dbfile))
```

```
drop table if exists projectmgr
```

```
drop table if exists projects
```

```
drop table if exists certifications
```

```
drop table if exists earnedcerts
```

```
create table projectmgr (  
  pmid integer,  
  name text not null,  
  primary key (pmid)  
)
```

```
create table certifications (  
  cid integer not null,  
  cert text not null,  
  body text not null,  
  primary key (cid)  
)
```

```
create table earnedcerts (  
  pmid integer,  
  cert integer not null,  
  primary key (pmid,cert)  
  foreign key (pmid) references projectmgr(pmid)  
  foreign key (cert) references certifications(cid)  
)
```

```
create table projects (
  pid integer,
  pname text default "No Name",
  budget double default 0,
  pmgr integer not null,
  primary key (pid),
  foreign key (pmgr) references projectmgr(pmid)
)
```

```
insert into certifications values
(1,"PMP","Project Management Institute"),
(2,"APM","Project Management Institute"),
(4,"CBAP","International Institute of Business Analysis"),
(6,"PSM","Scrum Organization"),
(5,"CIPM","American Academy of Project Management")
```

```
insert into projectmgr values
(11,"Xi Xufen"),
(22,"Ann Kelsey"),
(18,"Sandeep Ranghani"),
(19,"Alicia Knauss"),
(38,"Roxanna da Silva"),
(23,"Wagner Ormond"),
(47,"Margaret O'Shea"),
(43,"Kamala Patel"),
(20,"Alden White")
```

```
insert into earnedcerts values
(11,1),
(11,4),
(18,1),
(22,1),
(19,1),
(47,1),
(23,1),
(20,1),
(20,4),
(38,5)
```

```
insert into projects values
(100,"GOTAM",350000,11),
(200,"Data Twist",875000,20),
(300,"Alpha Launch",12500,18),
(302,"Alpha Arch Design",38000,18),
(303,"Alpha Dev",125000,18),
(305,"Alpha Analytics Dashboard",22900,38),
(505,"Training Portal Design",49900,19),
(811,"Bootstrap Demo",0,47)
```

```
select * from projectmgr
```

Table 1: 9 records

pmid	name
11	Xi Xufen
18	Sandeep Ranghani
19	Alicia Knauss
20	Alden White
22	Ann Kelsey
23	Wagner Ormond
38	Roxanna da Silva
43	Kamala Patel
47	Margaret O'Shea

```
select * from projects
```

Table 2: 8 records

pid	pname	budget	pmgr
100	GOTAM	350000	11
200	Data Twist	875000	20
300	Alpha Launch	12500	18
302	Alpha Arch Design	38000	18
303	Alpha Dev	125000	18
305	Alpha Analytics Dashboard	22900	38
505	Training Portal Design	49900	19
811	Bootstrap Demo	0	47

```
select * from certifications
```

Table 3: 5 records

cid	cert	body
1	PMP	Project Management Institute
2	APM	Project Management Institute
4	CBAP	International Institute of Business Analysis
5	CIPM	American Academy of Project Management
6	PSM	Scrum Organization

```
select * from earnedcerts
```

Table 4: Displaying records 1 - 10

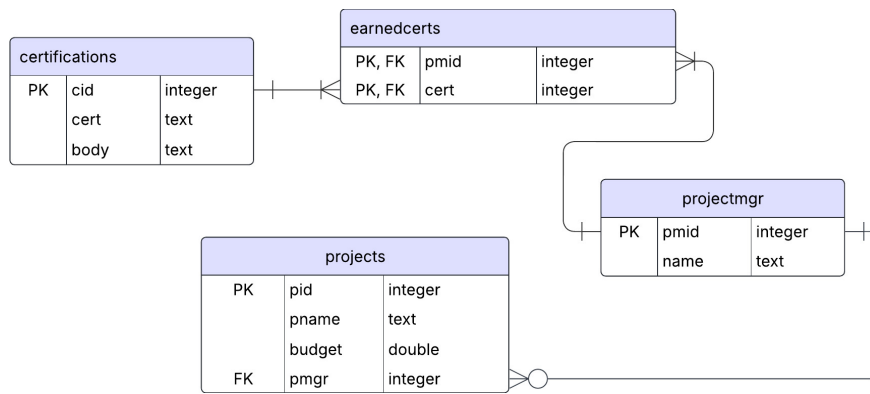
pmid	cert
11	1
11	4
18	1
22	1
19	1

pmid	cert
47	1
23	1
20	1
20	4
38	5

ERD of Project Database

The ERD for the project database is shown below:

My Lucid Chart is giving me difficulty to embed in a pdf with the provided HTML link from Lucid. Therefore I am embedding the image for the pdf as well as providing the link for HTML.



Q2.1 List the names of all project managers

```
SELECT DISTINCT name
FROM projectmgr
ORDER BY name;
```

Table 5: 9 records

name
Alden White
Alicia Knauss
Ann Kelsey
Kamala Patel
Margaret O'Shea
Roxanna da Silva
Sandeep Ranghani
Wagner Ormond
Xi Xufen

Q2.2 How many projects managers are there?

```
SELECT DISTINCT count(name) AS 'Number of Managers'
FROM projectmgr;
```

Table 6: 1 records

Number of Managers
9

Q2.3 List the names of all project managers who manage at least one project, ordered alphabetically starting with ‘A’

```
SELECT M.name, COUNT(P.pid) AS 'Project Count'
FROM projectmgr M JOIN projects P ON (M.pmid = P.pmgr)
GROUP BY M.name
HAVING COUNT(M.name) > 0
ORDER BY M.name;
```

Table 7: 6 records

name	Project Count
Alden White	1
Alicia Knauss	1
Margaret O'Shea	1
Roxanna da Silva	1
Sandeep Ranghani	3
Xi Xufen	1

Q2.4 List all project names, the name of the project manager, and the budget

```
SELECT P.pname AS 'Project Name', M.name AS 'Project Manager', P.budget AS 'Project Budget'
FROM projectmgr M JOIN projects P ON (M.pmid = P.pmgr)
ORDER BY P.pname;
```

Table 8: 8 records

Project Name	Project Manager	Project Budget
Alpha Analytics Dashboard	Roxanna da Silva	22900
Alpha Arch Design	Sandeep Ranghani	38000
Alpha Dev	Sandeep Ranghani	125000
Alpha Launch	Sandeep Ranghani	12500
Bootstrap Demo	Margaret O'Shea	0
Data Twist	Alden White	875000
GOTAM	Xi Xufen	350000
Training Portal Design	Alicia Knauss	49900

Q2.5 What is the average budget for all projects?

```
SELECT avg(budget) AS 'Average Budget Across All Projects'
FROM projects;
```

Table 9: 1 records

Average Budget Across All Projects
184162.5

Q2.6 Which project managers have more than two certifications? List their names. This query requires a GROUP BY

```
SELECT N.name, COUNT(C.cert) AS 'Certification Count'
FROM projectmgr N JOIN earnedcerts C ON (N.pmid = C.pmid)
GROUP BY N.name
HAVING COUNT(C.cert) > 2
ORDER BY N.name;
```

Table 10: 0 records

name	Certification Count

There are no managers with more than 2 certifications, if you change line 179 to ‘HAVING COUNT(C.cert) >= 2’ You will see that two managers (Alden White and Xi Xufen) have two certificates but no manager has above two.

Q2.7 How many projects have a budget that is below the average budget?

```
SELECT COUNT(pid) AS "Number of Projects below total average budget"
FROM projects
WHERE budget < (SELECT AVG(budget) FROM projects);
```

Table 11: 1 records

Number of Projects below total average budget
6

Q2. Which Projects have a project name that contains the text “Design” anywhere in the project name. List names of the projects

```
SELECT pname AS 'Project Name'
FROM projects
WHERE pname LIKE '%Design%'
```

Table 12: 2 records

Project Name
Alpha Arch Design
Training Portal Design

Q2.9 How many project managers manage projects that in total budget (i.e the sum of the budgets of their projects) are no more than \$50,000?

```
SELECT COUNT(*) AS 'Number of Project Managers'
FROM(
  SELECT M.name, SUM(P.budget) AS 'Total Budget'
  FROM projectmgr M JOIN projects P ON (M.pmid = P.pmgr)
  GROUP BY M.name
  HAVING SUM(P.budget) <= 50000
) AS subquery;
```

Table 13: 1 records

Number of Project Managers
3

Q2.10 Which project managers do not manage any projects? List their names

```
SELECT name
FROM projectmgr
WHERE pmid NOT IN (SELECT pmgr FROM projects)
ORDER BY name;
```

Table 14: 3 records

name
Ann Kelsey
Kamala Patel
Wagner Ormond

```
dbDisconnect(dbcon)
```